Seongwon (Gabriel) Yoon

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Research Interests

Optical Interconnects, Silicon Photonics, Network Architecture, AI Infrastructure

Education

Georgia Institute of Technology — Atlanta, GA, USA
Ph.D. in Electrical & Computer Engineering (PI: Prof. Shimeng Yu) — *Aug. 2024 – May. 2029 (expected)*

Seoul National University — Seoul, Korea
B.S. in Materials Science and Engineering, Summa Cum Laude (GPA 4.11/4.3) — *Mar. 2015 – Aug. 2022*

Korea Science Academy of KAIST — Busan, Korea
High School Diploma — *Mar. 2012 – Feb. 2015*

Experience

Georgia Institute of Technology — Atlanta, GA, USA
Graduate Research Assistant, Laboratory for Emerging Devices and Circuits — *Aug. 2024 – Present*

- Co-designed collective communication (all-reduce, all-to-all) with photonic switch configuration and GPU placement to run LLM traffic at full link rate.
- Laid out a silicon photonic MZI switch (SiEPIC PDK) and simulated it in Ansys Lumerical FDTD, MODE, HEAT, and INTERCONNECT.
- Extended the htsim packet-level simulator with new GPU fabrics and calibrated NVSwitch baselines for MoE training and decode.
- Built a full-package Ansys thermal model of a co-packaged optical GPU node and its laser, ring-tuning, and energy budgets.

Neuromorphic Materials and Devices Lab (NMDL) — Seoul, Korea
Undergraduate Researcher (PI: Prof. Sangbum Kim) — *Mar. 2019 – Oct. 2021, Jun. 2023 – Aug. 2024*

- 1.4M 6T2R phase change memory synaptic arrays with stochastic leaky-integrate-and-fire neurons simulation

Imagoworks Inc. — Seoul, Korea
Software Engineer, Deep CAD Team — *Jul. 2022 – Feb. 2023*

- Fine-tuned generative models for 3D image dental solutions

Samsung Electro-Mechanics — Suwon, Korea
Research Intern, Materials Research Group — *Jan. 2022 – Feb. 2022*

- Analyzed TEM data for MLCC ceramic dielectric

Bio-inspired Materials Lab — Seoul, Korea
Undergraduate Researcher (PI: Prof. Kitae Nam) — *Dec. 2018 – Feb. 2019*

- Fabricated carbon nanotube embedded Mg-based hydrogen generating catalysts

Publications

- **S. Yoon** and S. Yu, "An Optical KV Cache Network for Disaggregated LLM Serving," *arXiv preprint arXiv:2610.18444*, 2026.
- **S. Yoon**, P.-J. Chen, and S. Yu, "Cross-Layer Analysis of Thermal Tuning Stalls in Wafer-Scale Optical Interconnects for LLM MoE Training," *arXiv preprint arXiv:2608.24637*, 2026.
- F. Waqar, M.-Y. Lee, **S. Yoon**, S. Lim, and S. Yu, "CMOS+X: Stacking Memories Based on Oxide Transistors upon GPGPU Platforms," in *Proc. Int. Symp. Memory Syst. (MEMSYS)*, pp. 64–77, 2025.
- **S. Yoon**, J. Sharda, O. Phadke, and S. Yu, "A CMOS Image Sensor Pixel Readout Circuit Design Using Back-End-of-Line (BEOL)-Compatible Oxide Transistors," in *Proc. IEEE Int. Midwest Symp. Circuits Syst. (MWSCAS)*, pp. 1026–1030, 2025. **(Best Paper Award)**

- Y. G. Kang, M. Ishii, J. Park, U. Shin, S. Jang, **S. Yoon**, M. Kim, A. Okazaki, M. Ito, A. Nomura, et al., "Solving Max-Cut Problem Using Spiking Boltzmann Machine Based on Neuromorphic Hardware with Phase Change Memory," *Advanced Science*, vol. 11, no. 46, 2406433, 2024.
- J. Park, **S. Yoon**, H. Kim, Y. Kim, U. Lee, and H. Yu, "Clinical Validity and Precision of Deep Learning-Based Cone-Beam Computed Tomography Automatic Landmarking Algorithm," *Imaging Science in Dentistry*, vol. 54, no. 3, pp. 240–250, 2024.
- **S. Yoon**, U. Shin, and S. Kim, "Solving Constraint Satisfaction Problem with Spiking Neural Network Based on 1.4M 6T2R PCM Synaptic Array with 1.6K Stochastic LIF Neurons Neuromorphic Hardware," in *The 28th Korean Conference on Semiconductors (KCS)*, 2021.

Talks

Optical Interconnect for LLM, MoE and Agentic AI — Atlanta, GA, USA
Future of CMOS and Talent Workshop, Memory Wall and Interconnect Wall, Georgia Tech — *Mar. 2026*

Technical Skills

Photonic Simulation: Ansys Lumerical (FDTD, MODE, HEAT, INTERCONNECT), SiEPIC EBeam PDK

Device & Multiphysics: TCAD, Ansys Mechanical (thermal and structural)

Network Simulation: htsim, MixNet, FlexFlow

Circuit Design: schematic, layout, and tape-out in 180 nm CMOS

Programming: MATLAB

Selected Honors and Awards

Jung-hun Foundation Scholarship — South Korea
Full tuition support for 2 years — *2018 – 2020*

Eminence Scholarship, Seoul National University — South Korea
Full tuition support for 1.5 years — *2015 – 2016*